

Active v0.4.0, frozen. Deployed by explicit decision after two gated remediation attempts failed; trades 2021-2023-vintage development AUC for current-generator accuracy.

Task: real-versus-AI classification with a label, AI-positive score, CPU web/API/batch interface and model-influence explanation. Public-data results only; organizer data, baseline and hidden scores were not supplied.

Model, data and operating point

Frozen generic **CLIP ViT-L/14** image tower plus our trained **2-layer MLP head** (768->128->1, ReLU, dropout 0.3), replacing v0.3.0's linear head.
Training: **8,000 CIfAKE + 6,239 GenImage + 1,183 AIDA-train (17 current generators) + 2,122 CommunityForensics-train (~20 generators)**. 60/40 holdouts of both new sources fixed before training and never used for fitting.
Inference: EXIF orientation, official 224px CLIP centre crop, L2-normalized embedding, MLP head, temperature scaling. Temperature **0.226417**; AI threshold **0.838809**. Threshold uses internal GenImage/CIfAKE validation only; neither holdout was used to fit it.
Best-epoch selection (epoch 60/60) used internal validation AUC only; no holdout or external data influenced model/epoch selection.

Runnable interface and attribution

Shared CPU CLI/API; single image or batch with one loaded model. JPEG/PNG/WebP uploads up to **25 MiB / 40 MP**. EXIF is separate; C2PA presence is a heuristic substrng scan, not signature verification.
Tower **608,352,029 bytes** (reused unchanged from v0.3.0, parity verified) + head **398,527 bytes**, SHA-256 verified. **CPU-only** model runtime: torch, without clip/torchvision/ftfy/regex.
Measured CPU components (shared CLIP tower): predict **~0.38s/image**, explanation **~5.5s**, robustness **~5.1s**. Fresh public-clone setup+first-prediction: **243.75s** total, well under the 10-minute reproducibility bar.

Why this model exists: the gate failed twice, deployed by explicit decision

Two independent 2025-2026-generator evaluations on the previous release (v0.3.0): AI Detect Arena (17 generators) macro AUC **0.602**, accuracy **55.2%**, real-photo FPR **68.3%**. A model flagging two-thirds of real photos as AI is not usable. **Two gated remediation attempts both failed** on external-dev regression (linear+AIDA: 0.771->0.652 AUC; MLP+AIDA+CommunityForensics: 0.771->0.637 AUC), yet the MLP attempt's own untouched holdouts reached usable accuracy. Deployed by explicit user decision favouring current-generator accuracy over the old development benchmark - both facts disclosed together, not one substituted for the other.

Required metrics: overall AUC, unseen-generator-split AUC, macro-F1, confusion matrix

Evaluation surface	AUC	Macro-F1	Acc.	Confusion [real / ai_generated rows=actual]
Overall - GenImage val (seen, n=783)	0.944	0.738	74.7%	R:[366, 3] A:[195, 219]
Overall - CIfAKE val (seen, n=4000)	0.820	0.609	64.5%	R:[1900, 100] A:[1319, 681]
Unseen split - AI Detect Arena (17 generators, n=755)	0.948	0.852	85.2%	R:[329, 26] A:[86, 314]
Unseen split - CommunityForensics-Eval (~20 generators, n=1374)	0.936	0.717	71.8%	R:[539, 12] A:[375, 448]

Real->real / AI->AI: AIDA 92.7%/78.5%, CommunityForensics 97.8%/54.4%. Unseen-generator AUC (0.936-0.948) is not lower than seen-generator AUC (0.820-0.944): ranking generalises to new generators; the accuracy/FPR gap at the fixed threshold is where the deployment tradeoff below actually shows up. No organizer-provided baseline dataset or model exists for this internal hackathon; "Overall" here is our own in-distribution validation, not an organizer number. CommunityForensics-Eval is CC BY-NC-SA 4.0 (Park et al., CVPR 2025).

Cost: the 2021-2023-vintage development benchmark this release trades away

External dev (GLIDE-family generators)	v0.3.0	v0.4.0
Mean AUC, as distributed	0.771	0.637
Mean AUC, format matched	0.789	0.658

No GLIDE/DALLE or COCO reserved-set number is claimed for v0.4.0: that evaluation ran for v0.2.0/v0.3.0 only, and re-running it a third time was not judged worth spending that evidence on given the deployment decision already accepts this tradeoff. B-Free (third-party, CVPR 2025) scored 0.970/0.945 on the same 2021-2023-vintage development images without fitting - not our result.

Limitations

Explanation audit (re-run for this architecture): statistically indistinguishable from v0.3.0's linear head - identical 17/40 (42.5%) localisation count, deletion test not significant on either head. **Module G robustness:** real-photo FPR stays low (0-3.3%) under JPEG/resize/blur/screenshot, but AI recall drops 52.7%->17.9% under simulated screenshot; a bounded 7-transform search flips 33.5% of initially-correct predictions. **Private user-image veto check** (11 AI + 18 real, never used for fitting): real-photo false positives roughly halved versus v0.3.0. No generator attribution, caption consistency or C2PA signature verification. No organizer hidden-test score is available or claimed. Human explanation-usefulness review remains outstanding.

Sources: github.com/AI-Detect-Arena/benchmark-dataset; huggingface.co/datasets/OwensLab/CommunityForensics-Eval (Park et al., CVPR 2025); github.com/WisconsinAIVision/UniversalFakeDetect; github.com/openai/CLIP; github.com/GenImage-Dataset/GenImage; CIfAKE (Bird & Lotfi). Code and full experiments: github.com/sibtainmunshi/Signal_Scope.
Head SHA-256: 174ef56251011a7d5618d32759768807ebe02cc679c014e434906bbb81be5fdd